December 3, 1999

Inorganic Cume Organometallic Chemistry

1. Give a formula (or sketch) for a reasonably-common organic ligand which:

   a) can act as a one electron donor.
   b) can act as a two electron donor.
   c) can act as a three electron donor.

2. Give a formula (or sketch) for a reasonably-common organic ligand which:

   a) can act as a four electron donor.
   b) can act as a five electron donor.
   c) can act as a six electron donor.

3. Predict the most simple product of the high temperature and high pressure reaction of CO with

   a) Fe b) Mn c) Mo d) Ni

4. If the CO stretching frequency for $(PCl_3)_5Mo(CO)$ is 1989 cm^{-1}, will this number shift to a higher or lower value for the complex $(PPh_3)_5Mo(CO)$?

   Explain your reasoning.

5. Sketch two isomers of $Fe_2(CO)_9$.

   How can you experimentally determine which isomer you have?

6. The proton NMR of $Ru(C_8H_8)(CO)_3$ shows one peak at room temperature and 4 peaks at low temperature. Explain.

7. Why is $Ir(CO)(PPh_3)_2Cl$ often used in discussions of organometallic reactions?

8. Sketch likely structures for

   a) ferrocene b) $Co_3(CH)(CO)_9$
   c) $Fe_4(CO)_{13}^{-2}$ (hint: one Fe has a different environment from other 3)